Advanced Number Theory

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Divisibility rules and prime factorization

Find the prime factorization of 1260.

$1260 = 2^2 \times 3^2 \times 5 \times 7$.

2 [Foundational] • Target: Divisibility rules and prime factorization

Determine whether 143 is prime, and justify your answer.

$143 = 11 \times 13$, so 143 is not prime.

3 [Foundational] • Target: GCF, LCM, and their relationship

Find the GCF and LCM of 36 and 60.

$36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$. GCF $= 2^2 \times 3 = 12$. LCM $= 2^2 \times 3^2 \times 5 = 180$.

4 [Foundational] • Target: GCF, LCM, and their relationship

Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF × LCM = product of the two numbers.

GCF × LCM $= 8 \times 96 = 768$. Other number $= \frac{768}{24} = 32$.

5 [Foundational] • Target: Modular arithmetic

Find the remainder when 47 is divided by 6, and express this using modular notation.

$47 = 6(7) + 5$, so the remainder is 5. In modular notation: $47 \equiv 5 \pmod 6$.

6 [Foundational] • Target: Modular arithmetic

Find the last digit of 7^{100}, using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).

Since the cycle length is 4, find the remainder when 100 is divided by 4: $100 = 4(25) + 0$, so the remainder is 0, which corresponds to the 4th position in the cycle (7,9,3,1): the last digit is 1.

7 [Foundational] • Target: Sums and products of integer sets

Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.

$S = \frac{100(101)}{2} = 5050$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums and products of integer sets

Find the sum of all even integers from 2 to 200.

This is an arithmetic series with first term 2, last term 200, and $n = 100$ terms. Sum $= \frac{n(a_1 + a_n)}{2} = \frac{100(2 + 200)}{2} = \frac{100(202)}{2} = 10100$.

9 [Intermediate] • Target: Number sets and properties

Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.

$\sqrt{16} = 4$, a whole number, which is rational (it can be written as $\frac{4}{1}$). Even though it involves a square root symbol, the result simplifies to an integer.

10 [Intermediate] • Target: Number sets and properties

Explain why the sum of a rational number and an irrational number is always irrational.

If the sum of a rational number r and an irrational number x were rational (call it s), then $x = s - r$ would be the difference of two rational numbers, which is always rational — contradicting the assumption that x is irrational. So the sum must be irrational.

11 [Intermediate] • Target: Sequences and series

Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

$a_{15} = a_1 + (15 - 1)d = 6 + 14(4) = 6 + 56 = 62$.

12 [Intermediate] • Target: Sequences and series

Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, 54, \ldots$

First term $a_1 = 2$, ratio $r = 3$. Sum $= \frac{a_1(r^n - 1)}{r - 1} = \frac{2(3^8 - 1)}{2} = 3^8 - 1 = 6561 - 1 = 6560$.

13 [Intermediate] • Target: Absolute value and inequalities with integers

Find all integer solutions of $|x - 5| \le 3$.

$-3 \le x - 5 \le 3$, so $2 \le x \le 8$. Integer solutions: $2, 3, 4, 5, 6, 7, 8$.

© ClipSAT — clipsat.org | All Rights Reserved

14 **[Intermediate]** • **Target: Combining number theory concepts in a multi-part problem**

This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity GCF × LCM = product of the two numbers.

(a) $84 = 2^2 \times 3 \times 7$. $126 = 2 \times 3^2 \times 7$. (b) GCF (using lowest powers of shared primes): $2 \times 3 \times 7 = 42$. LCM (using highest powers of all primes present): $2^2 \times 3^2 \times 7 = 252$. (c) Verify:
GCF × LCM = $42 \times 252 = 10584$. Product of original numbers: $84 \times 126 = 10584$ ✓, confirming the identity holds.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Perfect squares, perfect cubes, and their properties** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

Find the smallest positive integer n such that $200n$ is a perfect square.

$200 = 2^3 \times 5^2$. For $200n$ to be a perfect square, every prime must appear to an even power. Currently 2^3 needs one more factor of 2 to become 2^4, and 5^2 is already even. So $n = 2$, giving $200(2) = 400 = 20^2$.

16 **[Challenging]** • **Target: Perfect squares, perfect cubes, and their properties** • **Cross-Lesson: 3.01 Advanced Functions**

Determine whether 128 is a perfect cube, and justify your answer using prime factorization.

$128 = 2^7$. For a perfect cube, the exponent must be divisible by 3. Since 7 is not divisible by 3, 128 is not a perfect cube.

17 **[Challenging]** • **Target: Counting divisors of a number** • **Cross-Lesson: 4.01 Advanced Trigonometry**

Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.

The number of divisors is found by adding 1 to each exponent and multiplying:
$(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 = 24$.

18 **[Challenging]** • **Target: Working with consecutive integers and integer constraints** • **Cross-Lesson: 5.01 Pre-Calculus Concepts**

Find three consecutive even integers whose sum is 96.

Let the integers be n, $n + 2$, $n + 4$: $n + (n + 2) + (n + 4) = 96$, so $3n + 6 = 96$, giving $3n = 90$, $n = 30$. The integers are 30, 32, 34.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The product of two consecutive positive integers is 156. Find the integers.

Let the integers be n and $n+1$: $n(n+1)=156$, so $n^2+n-156=0$. Factor: $(n+13)(n-12)=0$, giving $n=12$ (rejecting the negative solution). The integers are 12 and 13.

20 [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 7.01 Advanced Plane Geometry

A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.

Let the original number be $10t+u$ where t is the tens digit and u is the units digit: $t+u=11$. Reversed number: $10u+t$. Given: $(10u+t)-(10t+u)=27$, so $9u-9t=27$, giving $u-t=3$. Adding $t+u=11$ and $u-t=3$: $2u=14$, so $u=7$, then $t=4$. Original number: 47.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For what value of the digit d is the six-digit integer $452,d76$ divisible by 11?

a. 1

b. 3 ✓

c. 5

d. 7

Alternating sum test: $(6+d+5)-(7+2+4)=(11+d)-13=d-2$. For divisibility by 11, $d-2=1$ or 0. For $d-2+11k=0$, $d=2$ or with matching parity $d=3$ for $452,376$.

22 [Foundational] • Target: GCF, LCM, and their relationship

Find the remainder when 128 is divided by 9.

a. 1

b. 9

c. 2 ✓

d. 3

$128=9(14)+2$, so the remainder is 2.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Modular arithmetic

Find the remainder when 241 is divided by 12.

a. 0

b. 12

c. 1 ✓

d. 2

$241 = 12(20) + 1$, so the remainder is 1.

24 [Foundational] • Target: Sums and products of integer sets

Find the remainder when 78 is divided by 11.

a. 11

b. 0

c. 2

d. 1 ✓

$78 = 11(7) + 1$, so the remainder is 1.

25 [Foundational] • Target: Number sets and properties

Find the remainder when 205 is divided by 9.

a. 8

b. 9

c. 7 ✓

d. 6

$205 = 9(22) + 7$, so the remainder is 7.

26 [Foundational] • Target: Sequences and series

Find the remainder when 35 is divided by 11.

a. 2 ✓

b. 3

c. 1

d. 11

$35 = 11(3) + 2$, so the remainder is 2.

27 [Foundational] • Target: Absolute value and inequalities with integers

Find the remainder when 78 is divided by 10.

a. 7

b. 10

c. 9

d. 8 ✓

$78 = 10(7) + 8$, so the remainder is 8.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Combining number theory concepts in a multi-part problem

Find the remainder when 74 is divided by 13.

a. 8

b. 9 ✓

c. 10

d. 13

$74 = 13(5) + 9$, so the remainder is 9.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Perfect squares, perfect cubes, and their properties

Find the remainder when 177 is divided by 8.

a. 2

b. 1 ✓

c. 8

d. 0

$177 = 8(22) + 1$, so the remainder is 1.

30 [Intermediate] • Target: Counting divisors of a number

Find the LCM of 54 and 12.

a. 648

b. 6

c. 108 ✓

d. 114

$54 = 3^3 \times 2$, $12 = 3 \times 2^2$. LCM $= 3^3 \times 2^2 = 108$.

31 [Intermediate] • Target: Working with consecutive integers and integer constraints

Find the LCM of 12 and 18.

a. 6

b. 36 ✓

c. 216

d. 42

$12 = 2^2 \times 3$, $18 = 2 \times 3^2$. LCM $= 2^2 \times 3^2 = 36$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Digit sum and place value problems

Find the LCM of 24 and 18.

a. 432

b. 72 ✓

c. 78

d. 6

$24 = 2^3 \times 3$, $18 = 2 \times 3^2$. LCM $= 2^3 \times 3^2 = 72$.

33 [Intermediate] • Target: A multi-part modular arithmetic and remainder problem

Find the LCM of 375 and 135.

a. 50625

b. 3390

c. 15

d. 3375 ✓

$375 = 5^3 \times 3$, $135 = 5 \times 3^3$. LCM $= 5^3 \times 3^3 = 3375$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the LCM of 100 and 250.

a. 500 ✓

b. 25000

c. 50

d. 550

$100 = 2^2 \times 5^2$, $250 = 2 \times 5^3$. LCM $= 2^2 \times 5^3 = 500$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the LCM of 250 and 200.

a. 1050

b. 50000

c. 1000 ✓

d. 50

$250 = 5^3 \times 2$, $200 = 5^2 \times 2^3$. LCM $= 5^3 \times 2^3 = 1000$.

36 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the LCM of 225 and 135.

a. 675 ✓

b. 30375

c. 45

d. 720

$225 = 5^2 \times 3^2$, $135 = 5 \times 3^3$. LCM $= 5^2 \times 3^3 = 675$.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the 14th term of the arithmetic sequence with first term -4 and common difference 3.

a. 38

b. 32

c. 37

d. 35 ✓

$a_{14} = a_1 + (14 - 1)d = -4 + 39 = 35$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the 7th term of the geometric sequence $2, 4, 8, 16, \ldots$

a. 130

b. 125

c. 131

d. 128 ✓

First term $a_1 = 2$, ratio $r = 2$. $a_7 = a_1 \cdot r^{7-1} = 2 \cdot 2^6 = 128$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the 5th term of the geometric sequence $1, 2, 4, 8, \ldots$

a. 19

b. 13

c. 18

d. 16 ✓

First term $a_1 = 1$, ratio $r = 2$. $a_5 = a_1 \cdot r^{5-1} = 1 \cdot 2^4 = 16$.

40 [Challenging] • Target: Divisibility rules and prime factorization • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the number of positive divisors of 528, using its prime factorization $528 = 2^4 \times 3 \times 11$.

a. 18

b. 20 ✓

c. 22

d. 4

Add 1 to each exponent and multiply: $(4 + 1)(1 + 1)(1 + 1) = 5 \times 2 \times 2 = 20$.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: GCF, LCM, and their relationship • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the number of positive divisors of 900, using its prime factorization $900 = 2^2 \times 3^2 \times 5^2$.

a. 25

b. 8

c. 29

d. 27 ✓

Add 1 to each exponent and multiply: $(2 + 1)(2 + 1)(2 + 1) = 3 \times 3 \times 3 = 27$.

42 [Challenging] • Target: Modular arithmetic • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the last digit of 3^{24}, using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).

a. 7

b. 3

c. 1 ✓

d. 9

Since the cycle length is 4, find the remainder when 24 is divided by 4: $24 = 4(6)$, so the remainder is 0, corresponding to position 4 in the cycle: the last digit is 1.

43 [Challenging] • Target: Sums and products of integer sets • Cross-Lesson: 3.01 Advanced Functions

Find the last digit of 3^{132}, using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).

a. 7

b. 9

c. 1 ✓

d. 3

Since the cycle length is 4, find the remainder when 132 is divided by 4: $132 = 4(33)$, so the remainder is 0, corresponding to position 4 in the cycle: the last digit is 1.

44 [Challenging] • Target: Number sets and properties • Cross-Lesson: 4.01 Advanced Trigonometry

Find the last digit of 4^{129}, using the pattern of the last digits of powers of 4 (which cycle every 2: 4, 6).

a. 4 ✓

b. 6

c. 7

d. 9

Since the cycle length is 2, find the remainder when 129 is divided by 2: $129 = 2(64) + 1$, so the remainder is 1, corresponding to position 1 in the cycle: the last digit is 4.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

What is the units digit of the integer 7^{2024}?

a. 1 ✓

b. 3

c. 7

d. 9

Units digits of powers of 7 repeat in a cycle of length 4: $7, 9, 3, 1$. Since 2024 is divisible by 4, the units digit is 1.

© ClipSAT — clipsat.org | All Rights Reserved